

ARTHAT 5.0 - CASE STUDY COMPETITION

TEAM CASE WIZARDS

(SRI GURU GOBIND SINGH COLLEGE OF COMMERCE)

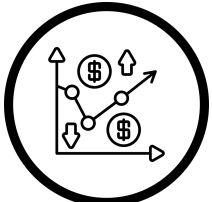
MEHA SINGHAL

MILAN KUMAR

EXECUTIVE SUMMARY



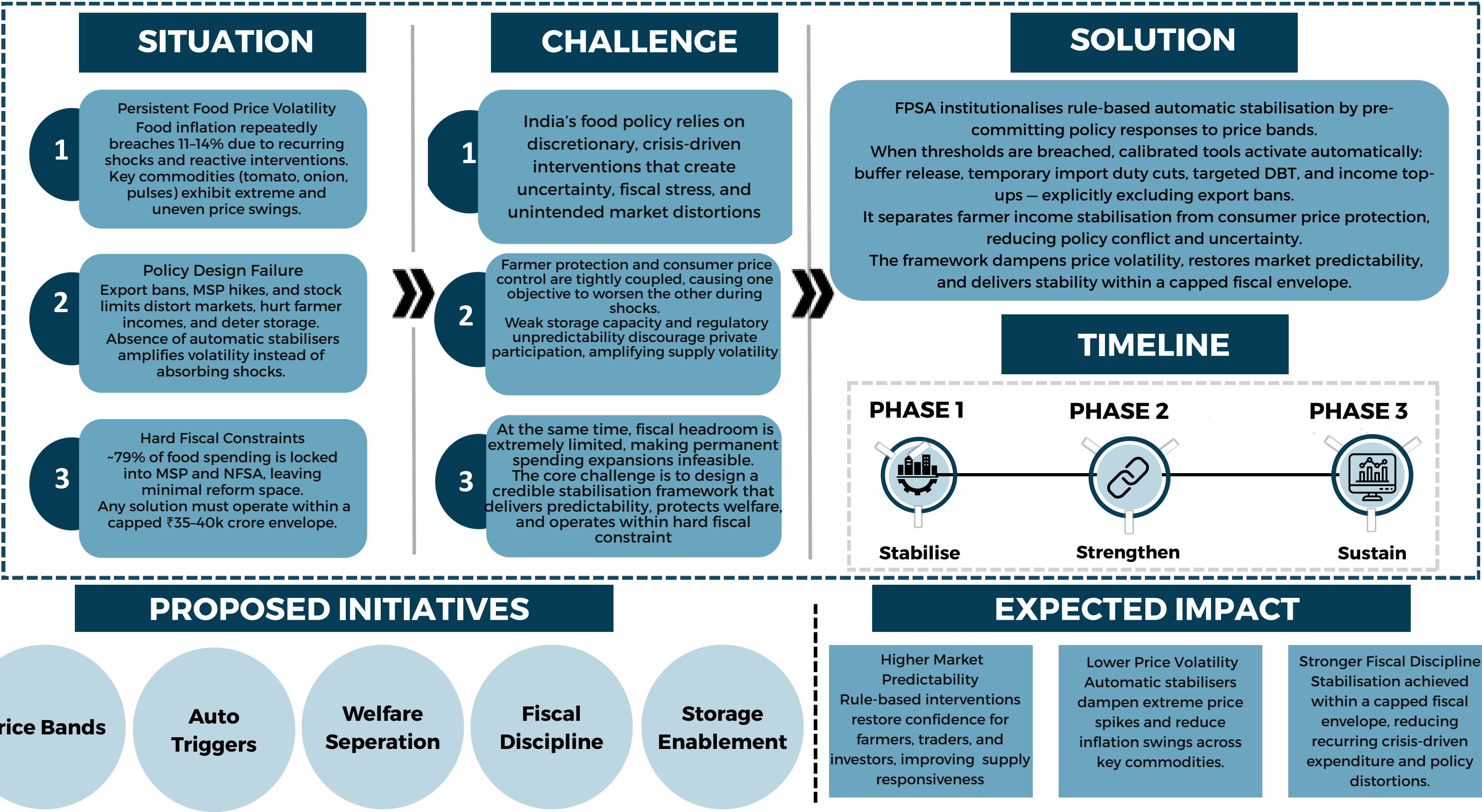
2.1 LAKH
Food policy already absorbs ₹2.1 lakh cr (NFSA) + ₹1.4-1.6 lakh cr (MSP). Any solution must operate within a ₹35,000-40,000 cr cap.



PRICE VOLATILITY
Surplus procurement, export bans, and emergency interventions create hidden fiscal leakages that compound over time.



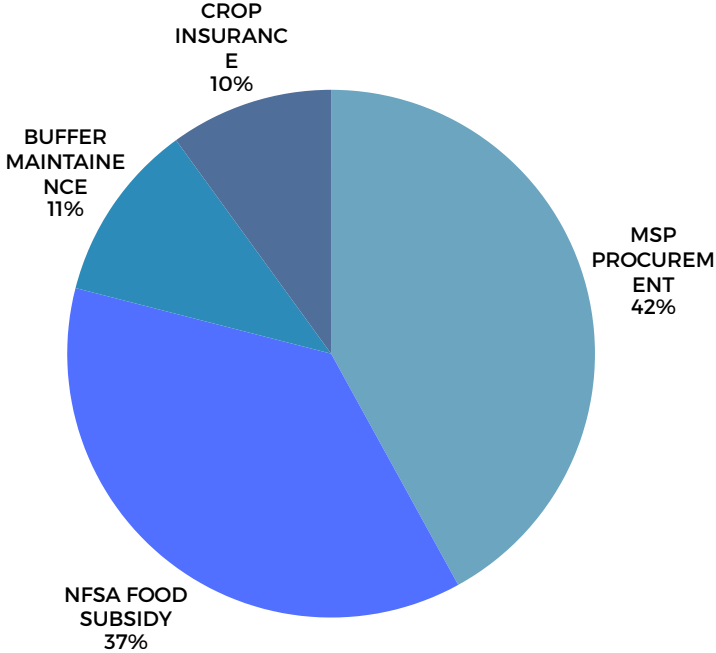
TIME BOUND
Universal subsidies, unlimited MSP escalation, and permanent schemes would lock in future deficits and breach fiscal limits.



Why Food Inflation Keeps Spiralling: A Policy Design Failure

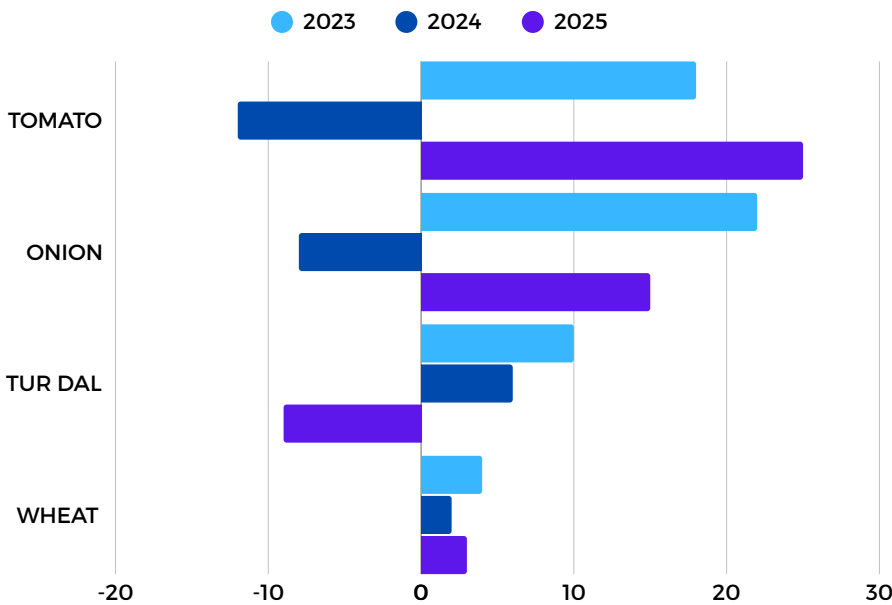
India's food inflation problem is structural, driven by recurring policy responses rather than isolated shocks.

FISCAL BURDEN OF FOOD POLICIES



79% of total spending is locked into MSP procurement + NFSA subsidy, leaving very limited fiscal flexibility. Buffer maintenance and crop insurance together form only ~21%, despite being critical for volatility management and climate risk

PRICE VOLATILITY ACROSS COMMODITIES



Observed price volatility patterns reflect underlying market and policy asymmetries, justifying differentiated food-price and agricultural policy interventions rather than a uniform approach.

1

Income Stabilisation over Price Control
Use price-deficiency payments and targeted support instead of blanket MSP hikes and bans.

2

Automatic Shock Absorbers
Rule-based buffer release and trade triggers linked to price bands.

3

Commodity-Specific Policy Design
Stabilisation tools for vegetables & pulses; calibrated procurement for cereals.

4

Storage & Supply Chain Strengthening
Expand cold storage, processing, and private warehousing to smooth supply.

5

Predictable Trade & Regulation
Stable tariffs and reduced discretionary controls to restore market confidence.

MONSOON FAILURE



EXPORT BAN



FARMERS PRICE CRASH



LOWER FUTURE PRODUCTION



NEXT PRICE SPIKE

SYSTEM FAILURE

- No automatic stabilisers (every shock needs fresh decisions)
- Weak storage & logistics (only ~10% perishables in cold storage)
- Import exposure without buffers
- Trader uncertainty due to raids & stock limits

IMPACT ON STAKEHOLDERS

- Farmers: Income collapses due to sudden policy interventions.
- Consumers: Face sharp, unpredictable food price spikes.
- Government: Bears rising fiscal costs and credibility loss.
- Traders: Avoid storage due to regulatory fear.

Food Price Shock Absorber (FPSA): From Discretion to Rules

FPSA institutionalises automatic stabilisation in food markets by pre-committing policy responses to price movements, rather than reacting ex-post through discretionary measures.

1. RULE-BASED COMMITMENT REDUCES POLICY VOLATILITY

Between 2023–25, food inflation frequently breached 11–14%, far above RBI’s 4% target
Each episode triggered different tools (export bans, MSP hikes, stock limits)

FPSA change:

One Predefined response → Lower Uncertainty



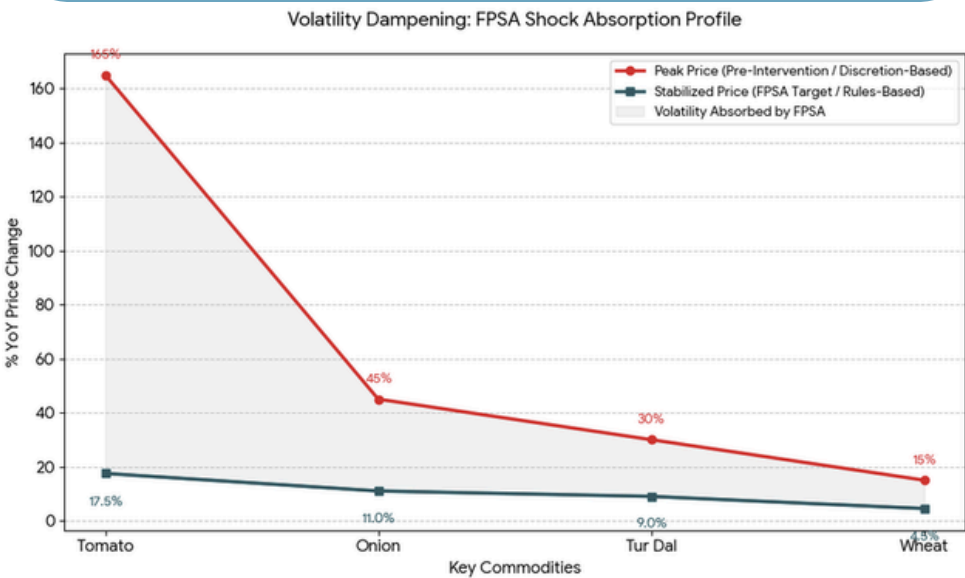
2. FOCUS ON VOLATILITY NOT PRICE SUPPRESSION

- Tomato prices touched ₹200/kg (+165% YoY)
- Onion prices rose to ₹90/kg (+45% YoY)* before policy intervention
- FPSA does *not* aim to keep prices low, but to *prevent extreme deviations.

Target: Narrow price swings, not fix prices.

3. PREDICTABILITY AS AN ECONOMIC STABILISER

- Traders face raids and stock limits → reduce storage
 - Only ~10% of perishables have cold storage access
- FPSA predictability incentivises private storage, easing supply shocks without new public infrastructure upfront.



4. FISCAL DISCIPLINE EMBEDDED BY DESIGN

- MSP procurement already costs ₹1.4–1.6 lakh crore annually
 - NFSA food subsidy is ₹2.1 lakh crore
- FPSA avoids adding permanent commitments to an already stretched base.

5. SEPARATION OF WELFARE OBJECTIVES

Under current policy:

- Farmer protection relies heavily on MSP hikes
- Consumer protection relies on export bans
- Fiscal cost balloons as both operate simultaneously

FPSA separation:

Farmer welfare → income stabilisation
Consumer welfare → targeted protection
Volatile Environment → volatility containment

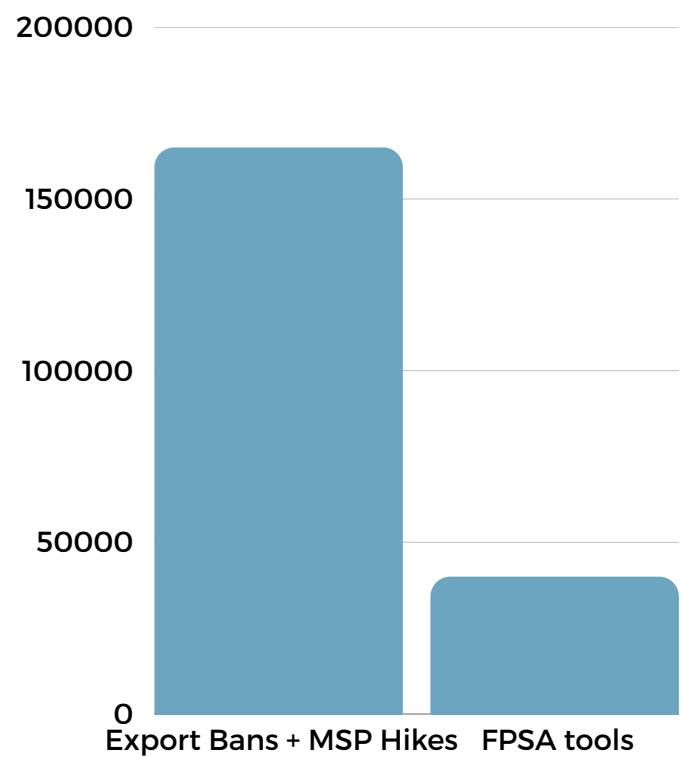
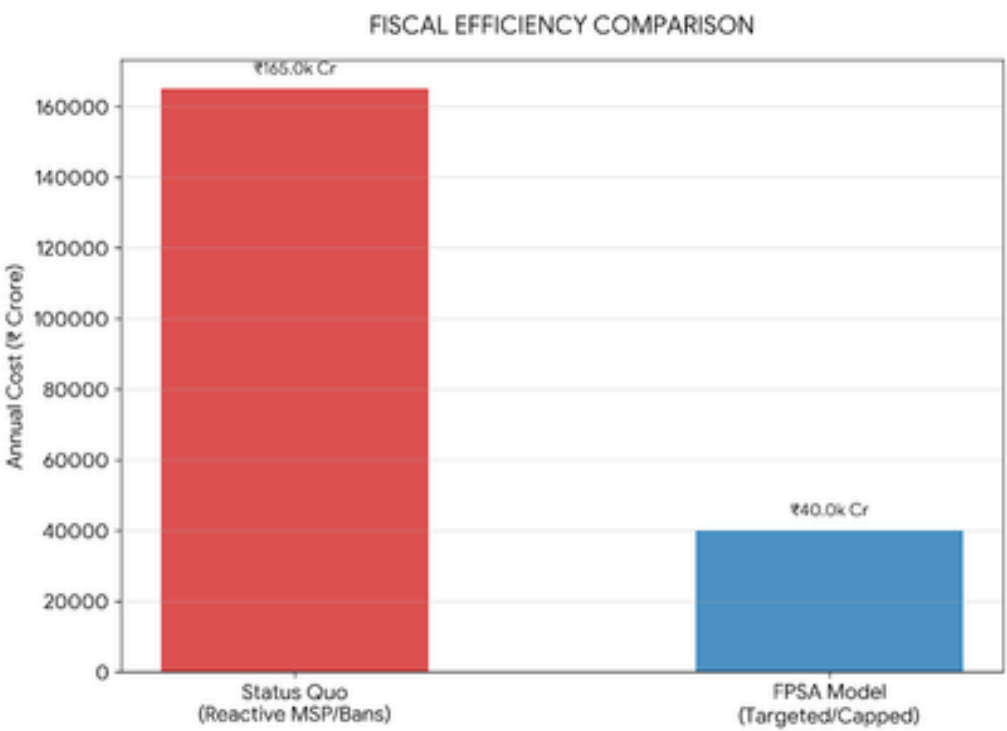
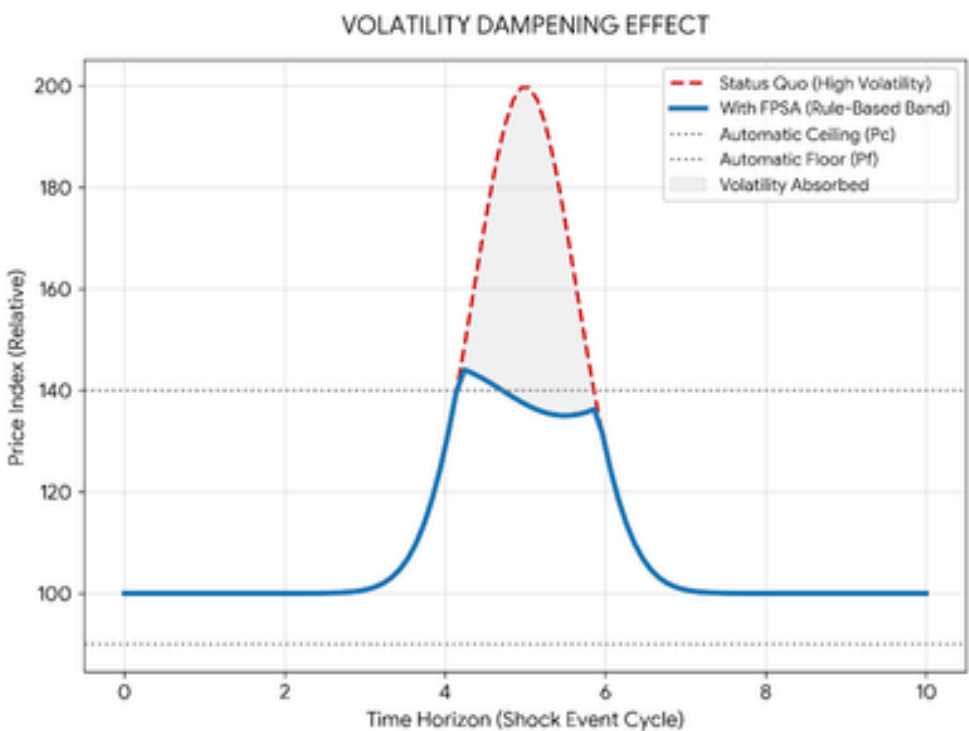
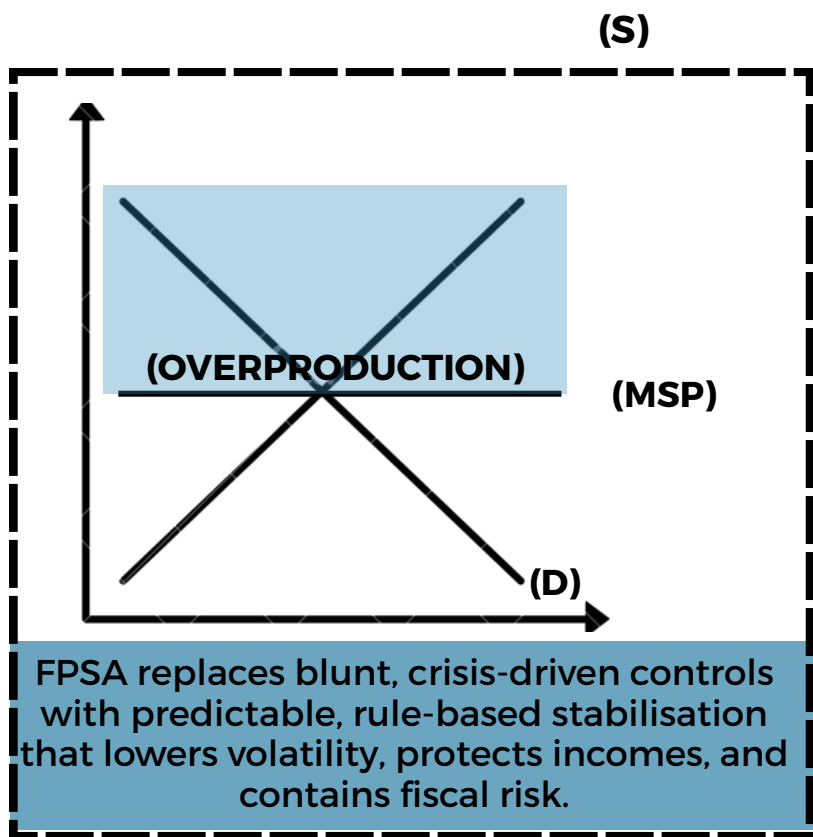
Discretion vs Rules — Conceptual Comparison Table				
Dimension	Discretion-Based	Rule-Based (FPSA)	Rule-Based (FPSA)	Comparison
Predictability	Low	High	High	↑
Speed	Delayed	Immediate	Immediate	↓
Political Risk	High	Low	Low	↓
Market Confidence		Weak	Strong	↑

FPSA DELIBERATELY EXCLUDES

- No permanent price controls
- No universal subsidies
- No open-ended procurement
- No politically timed interventions

Rapid Stabilisation: Automatic Price Bands for Critical Foods

A rule-based system that dampens food price shocks, protects farmers and consumers, and stabilises inflation—without bans, distortions, or fiscal drift.



Rule-based FPSA delivers price stability at nearly ¼ the fiscal cost of reactive export bans and MSP hikes.

AUTOMATIC TRIGGER RESPONSE SYSTEM

If price > Pc:
Buffer stock release, temporary import duty cut to 0% (≤90 days), targeted DBT to bottom 40% households.

If price < Pf:
Direct income support to farmers, quantity-capped procurement, diversion to processing.
Export bans are explicitly ruled out.

SUPPORTING DESIGN SAFEGUARDS

Pf and Pc are formula-driven using rolling cost and volatility averages, limiting discretion and anchoring market expectations.

Bands differ across cereals, pulses, oilseeds, and perishables based on storability and import elasticity, improving targeting.

All actions operate within a fixed FPSA envelope, with automatic throttling once limits are reached.

Weekly disclosures and independent audits ensure credibility, while interventions unwind automatically once prices normalise.

STAKEHOLDER

Farmers	reduced income volatility, predictable support.
Consumers	fewer price spikes, smoother inflation.
Government	lower fiscal risk and higher policy credibility.
Traders	greater regulatory certainty, improved storage incentives.
Importers	predictable trade policy reduces shock transmission.

OUTCOME OF THE PLAN

Outcome	Direction
Volatility	↓ ↓ ↓
Predictability	↑ ↑ ↑
Fiscal risk	↓ ↓

Fiscal Discipline and Inevitable Trade-offs

FPSA stabilises food inflation within a hard fiscal envelope by reallocating spending and accepting unavoidable trade-offs.

HARD FISCAL CONSTRAINTS

Maximum additional fiscal space: ₹35,000–40,000 crore

Food subsidy (NFSA) base: ₹2.1 lakh crore

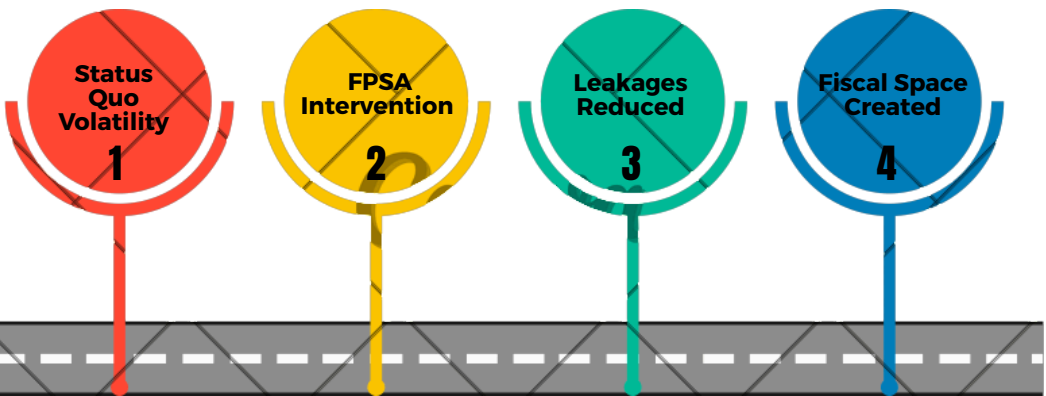
NFSA expansion cap: ≤10%

MSP procurement already costs ₹1.4–1.6 lakh crore annually

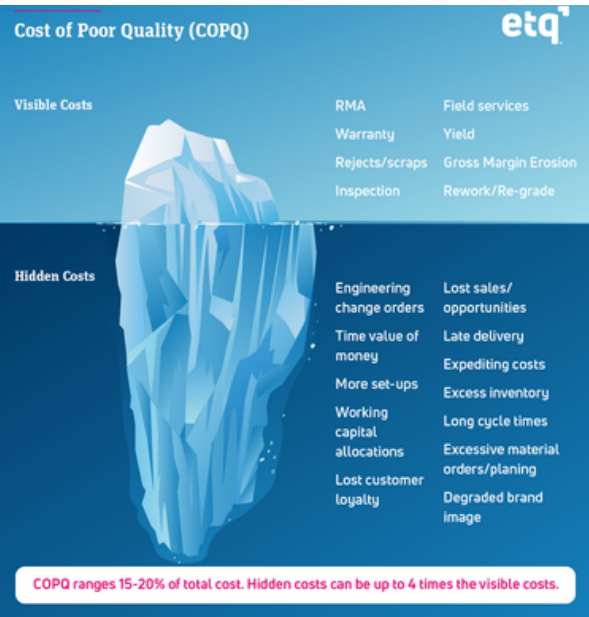
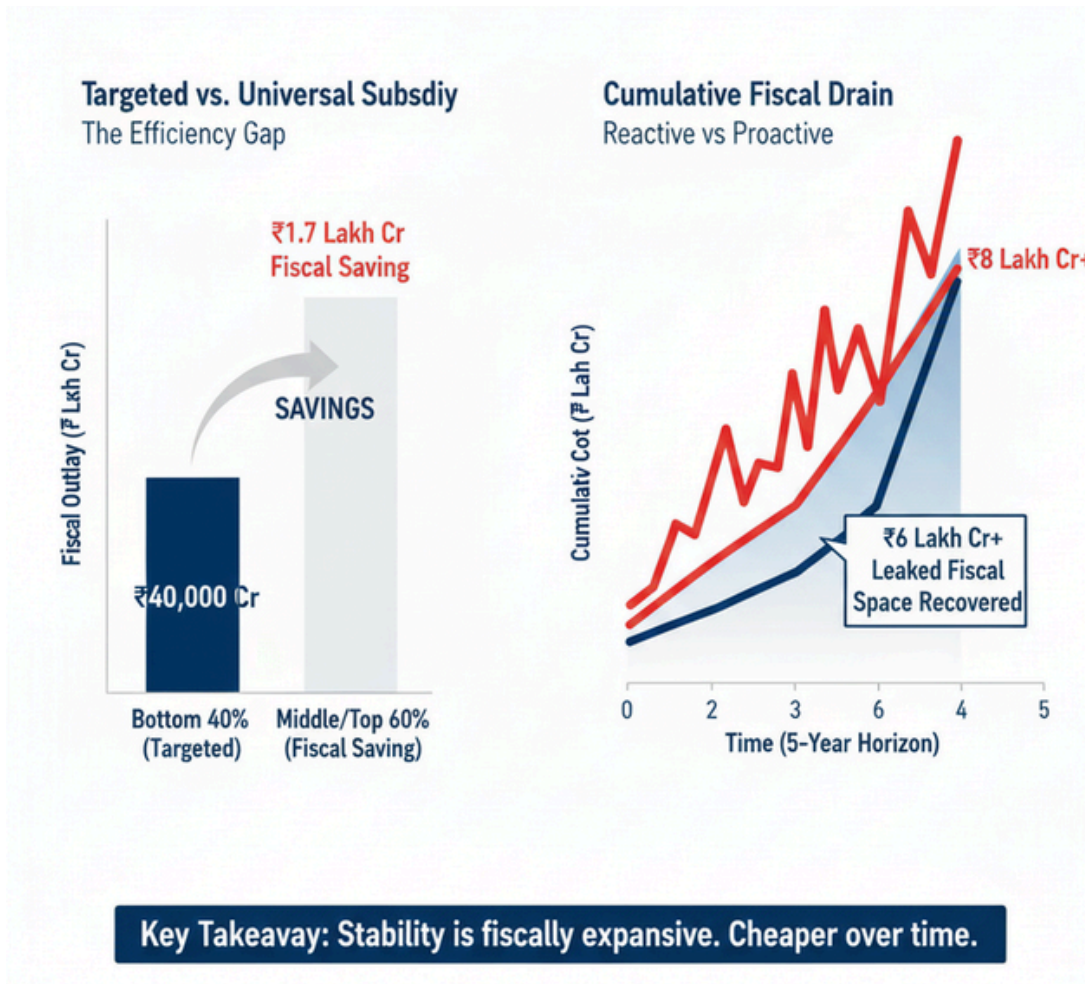
Buffer stock maintenance costs ~₹11,000 crore/year

Implication:
Any solution that adds permanent spending is infeasible.

FPSA FISCAL ROADMAP



INSIGHT
Volatility is fiscally expensive. Stability is cheaper over time.



Poor quality doesn't just create defects—it creates systemic, long-term financial damage. Managing visible issues without addressing hidden costs gives a false sense of control.



INEVITABLE TRADE-OFFS

- 1 No universal DBT means middle-income households face market price spikes, but limiting support to the bottom 40% avoids breaching the ₹2.1 lakh crore allocation and NFSA cap while preserving fiscal credibility.
- 2 MSP is not used for price stabilisation since procurement already costs ₹1.4–1.6 lakh crore annually, and further hikes would create surpluses, storage losses, and fiscal drag.
- 3 Farmers lose artificial price floors during extreme shortages, but past export bans led to around 35% onion price crashes, ultimately hurting farmers more than price volatility itself.
- 4 There are no open-ended schemes or permanent entitlement expansions, as such measures would lock in future deficits; instead, FPSA tools are time-bound and reversible.

WHY THESE TRADE-OFFS ARE RATIONALLY OPTIMAL
The choice is not between pain and no pain – it is between visible, controlled pain and hidden, compounding fiscal damage

STAKEHOLDER IMPACT

Government:
Fiscal control + policy credibility

Farmers:
Income stability without surplus collapse

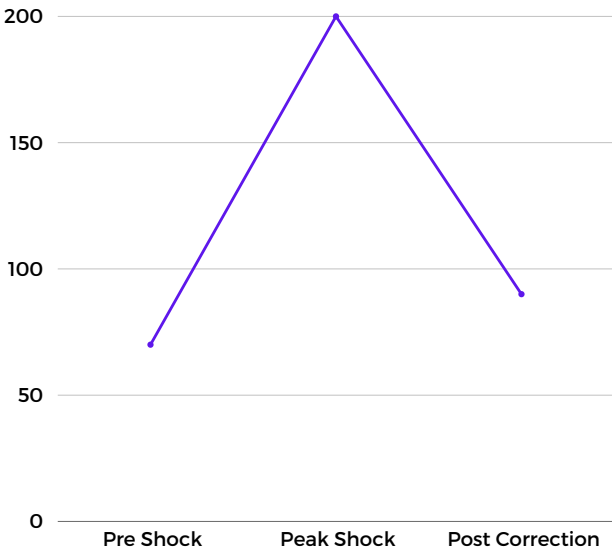
Consumers:
Protection focused on most vulnerable

Taxpayers:
Avoids permanent subsidy expansion

Building a Low-Volatility Food Economy

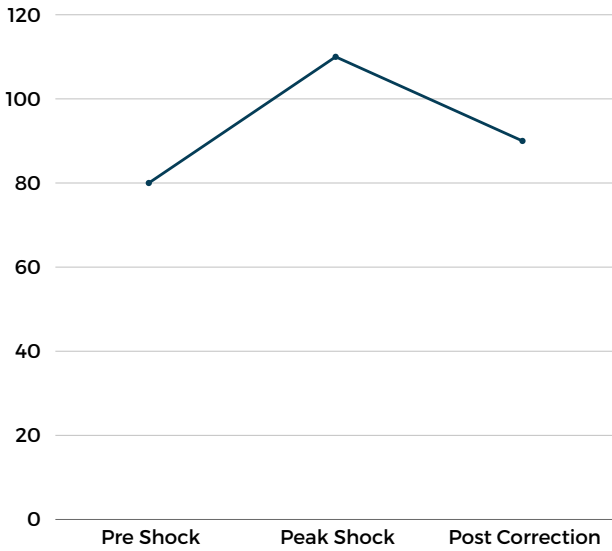
FPSA reduces volatility in the short run; structural reforms ensure future shocks are smaller, rarer, and cheaper to manage.

High Volatility Scenario



- Extreme spike
- Large amplitude
- Slow, uneven correction

Low Volatility Scenario



- Controlled seasonal movement
- No extreme spike
- Faster normalization

How Volatility Reduction Lowers Fiscal Pressure

Today (High Volatility)

- Repeated interventions
- MSP escalation
- Export bans → compensation & distortions
- Rising baseline subsidies
- (NFSA: ~₹2.1 lakh crore)

Future (FPSA + Reforms)

- Fewer shock episodes
- Lower intervention frequency
- Targeted, time-bound support
- Spending becomes episodic, not permanent

FOUR STRUCTURAL PILLARS

From Price Protection → Income Stability

Decoupling farmer income security from price support prevents overproduction, lowers surplus cycles, and slows the fiscal growth of MSP procurement.

Private Storage as First Line of Defence

Replacing stock limits with disclosure unlocks cheaper private storage, smoothing seasonal supply and reducing repeated government buffer interventions.

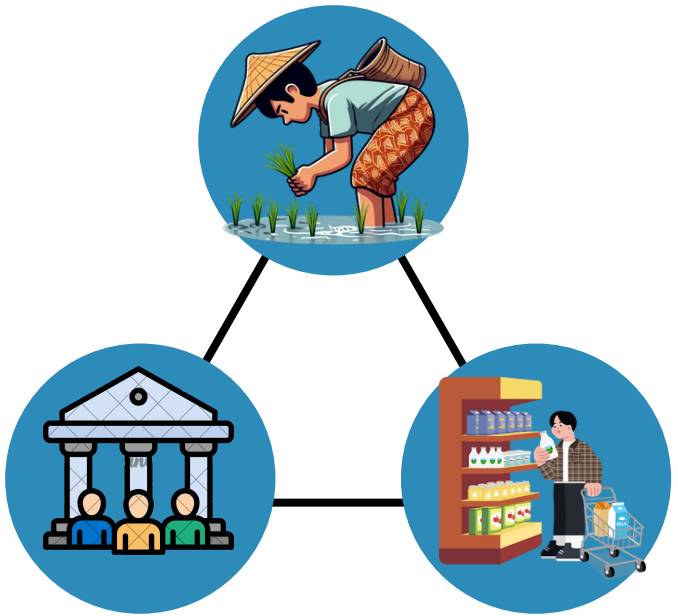
Infrastructure Linked to Inflation Outcomes

Tying cold-chain and irrigation spending to local inflation and loss metrics converts capex into measurable price stability rather than sunk investment.

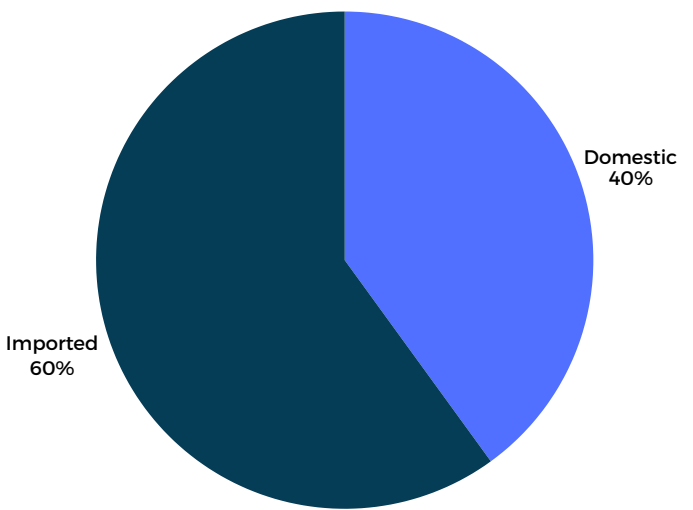
Reduce External Shock Transmission (Edible Oils)

Strategic buffers, long-term imports, and gradual domestic production dampen global price pass-through and stabilise edible oil inflation.

System Stakeholder Stability Triangle



Edible Oil Import Dependence

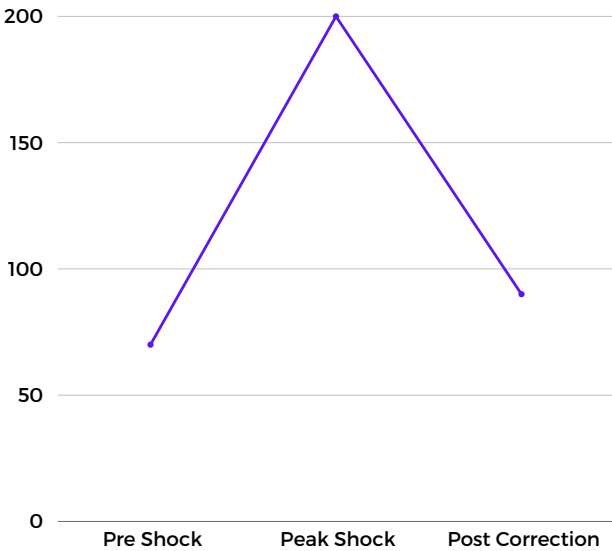


High import dependence converts global price swings into unavoidable domestic inflation.

Building a Low-Volatility Food Economy

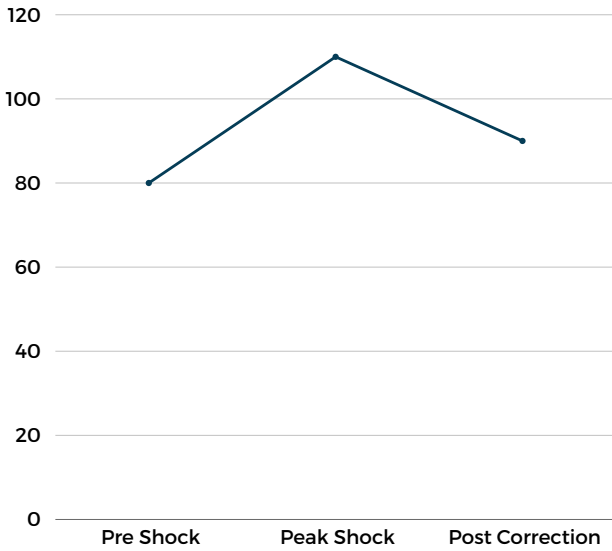
FPSA reduces volatility in the short run; structural reforms ensure future shocks are smaller, rarer, and cheaper to manage.

High Volatility Scenario



- Extreme spike
- Large amplitude
- Slow, uneven correction

Low Volatility Scenario



- Controlled seasonal movement
- No extreme spike
- Faster normalization

How Volatility Reduction Lowers Fiscal Pressure

Today (High Volatility)

- Repeated interventions
- MSP escalation
- Export bans → compensation & distortions
- Rising baseline subsidies
- (NFSA: ~₹2.1 lakh crore)

Future (FPSA + Reforms)

- Fewer shock episodes
- Lower intervention frequency
- Targeted, time-bound support
- Spending becomes episodic, not permanent

FOUR STRUCTURAL PILLARS

From Price Protection → Income Stability

Decoupling farmer income security from price support prevents overproduction, lowers surplus cycles, and slows the fiscal growth of MSP procurement.

Private Storage as First Line of Defence

Replacing stock limits with disclosure unlocks cheaper private storage, smoothing seasonal supply and reducing repeated government buffer interventions.

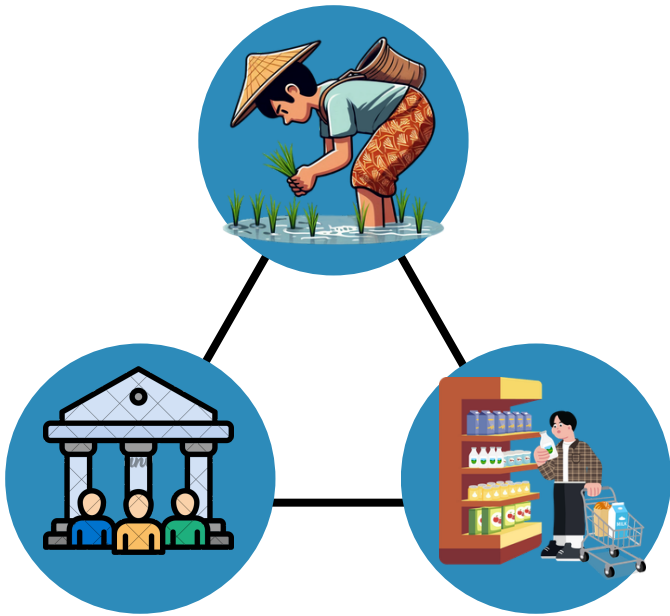
Infrastructure Linked to Inflation Outcomes

Tying cold-chain and irrigation spending to local inflation and loss metrics converts capex into measurable price stability rather than sunk investment.

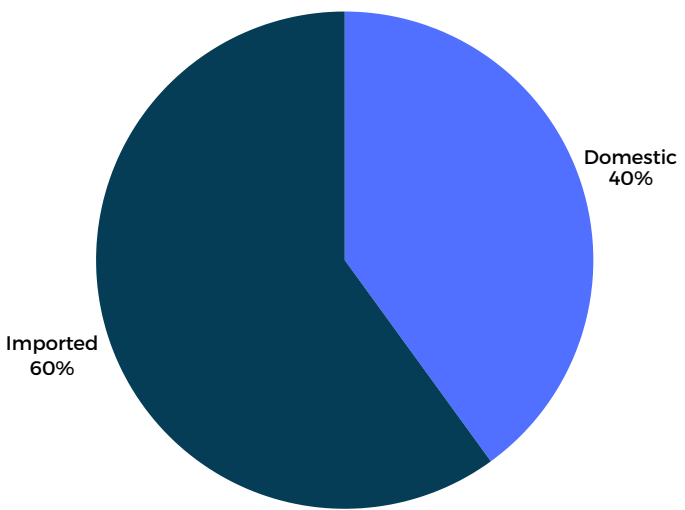
Reduce External Shock Transmission (Edible Oils)

Strategic buffers, long-term imports, and gradual domestic production dampen global price pass-through and stabilise edible oil inflation.

System Stakeholder Stability Triangle



Edible Oil Import Dependence

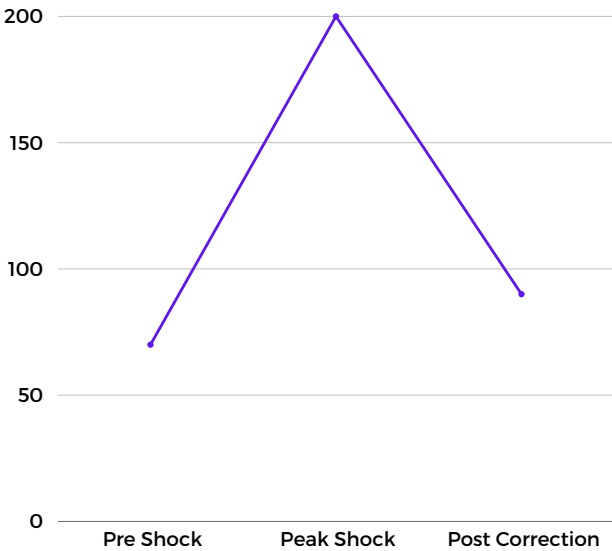


High import dependence converts global price swings into unavoidable domestic inflation.

Building a Low-Volatility Food Economy

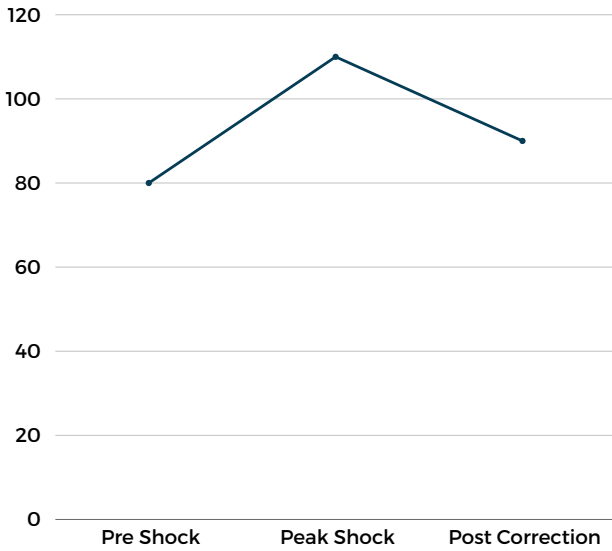
FPSA reduces volatility in the short run; structural reforms ensure future shocks are smaller, rarer, and cheaper to manage.

High Volatility Scenario



- Extreme spike
- Large amplitude
- Slow, uneven correction

Low Volatility Scenario



- Controlled seasonal movement
- No extreme spike
- Faster normalization

How Volatility Reduction Lowers Fiscal Pressure

Today (High Volatility)

- Repeated interventions
- MSP escalation
- Export bans → compensation & distortions
- Rising baseline subsidies
- (NFSA: ~₹2.1 lakh crore)

Future (FPSA + Reforms)

- Fewer shock episodes
- Lower intervention frequency
- Targeted, time-bound support
- Spending becomes episodic, not permanent

FOUR STRUCTURAL PILLARS

From Price Protection → Income Stability

Decoupling farmer income security from price support prevents overproduction, lowers surplus cycles, and slows the fiscal growth of MSP procurement.

Private Storage as First Line of Defence

Replacing stock limits with disclosure unlocks cheaper private storage, smoothing seasonal supply and reducing repeated government buffer interventions.

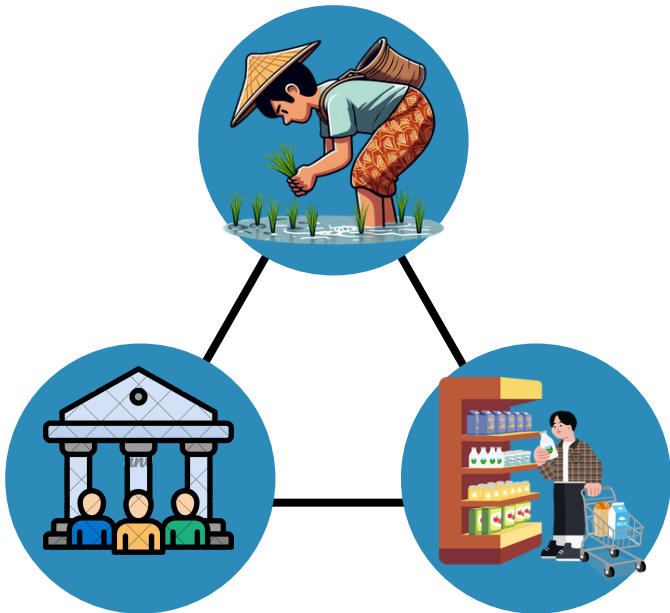
Infrastructure Linked to Inflation Outcomes

Tying cold-chain and irrigation spending to local inflation and loss metrics converts capex into measurable price stability rather than sunk investment.

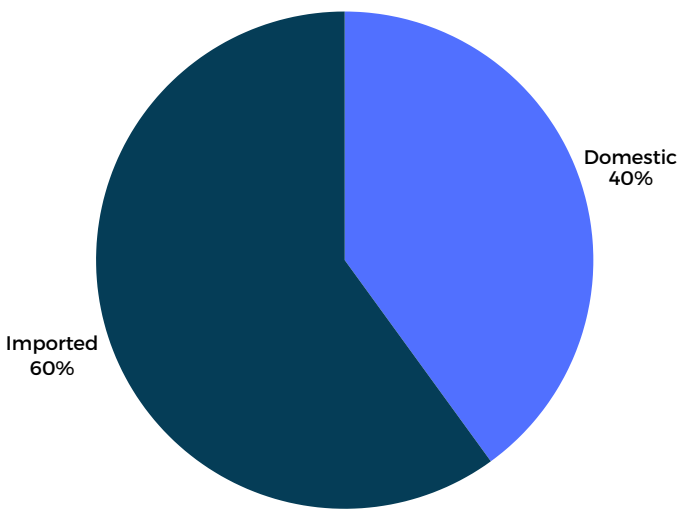
Reduce External Shock Transmission (Edible Oils)

Strategic buffers, long-term imports, and gradual domestic production dampen global price pass-through and stabilise edible oil inflation.

System Stakeholder Stability Triangle



Edible Oil Import Dependence



High import dependence converts global price swings into unavoidable domestic inflation.

THANK YOU

TEAM CASE WIZARDS

(SRI GURU GOBIND SINGH COLLEGE OF COMMERCE)

MEHA SINGHAL | meha.242106@sggscs.ac.in | +91 88268 04636

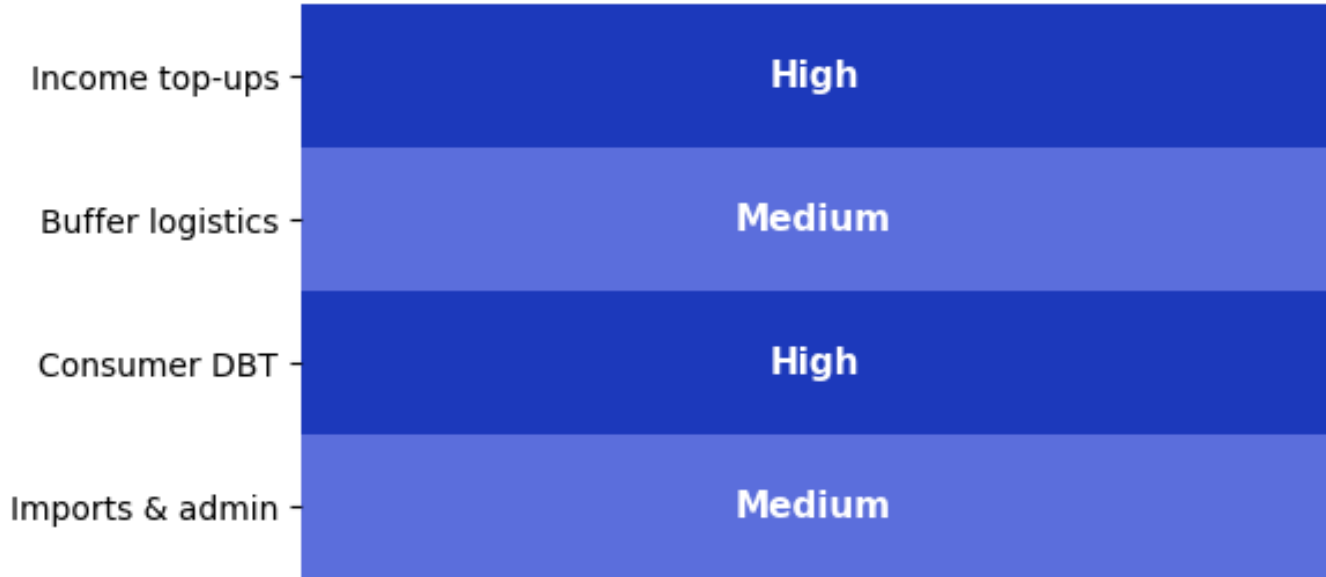
MILAN KUMAR | ismilan2004@gmail.com | +91 98718 74283

APPENDIX

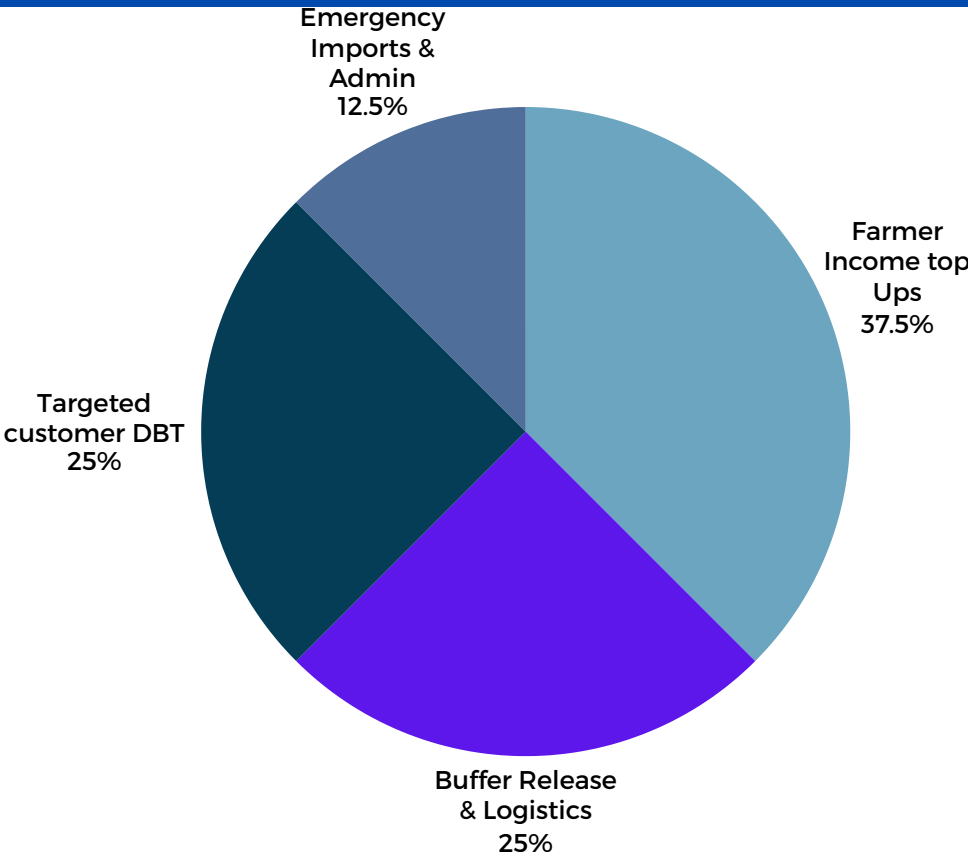
Annual Fiscal Outlay Under FPSA

FPSA Component	Annual Outlay (₹ crore)	Nature
Farmer income top-ups	~15,000	Variable, shock-linked
Buffer release & logistics	~10,000	Temporary
Targeted consumer DBT (bottom 40%)	~10,000	Shock-period only
Emergency imports & admin	~5,000	Contingent
Total	~40,000	Capped

Fiscal Flexibility Heat Map (FPSA)



FPSA Spending – Time-Bound Nature (₹40,000 cr Total)



FPSA is designed to respond to volatility, not institutionalise it.

- High-flexibility components (income top-ups, consumer DBT) absorb shocks without creating permanent fiscal liabilities.
- Medium-flexibility items (logistics, imports) are operationally constrained but still time-bound, preventing baseline creep.

100% of FPSA spending is temporary, shock-linked, or contingent—zero permanent entitlements.

1. The largest share (farmer income top-ups) activates only when prices deviate sharply, not as an annual payout.



2. Consumer protection is targeted and episodic, avoiding blanket subsidies.



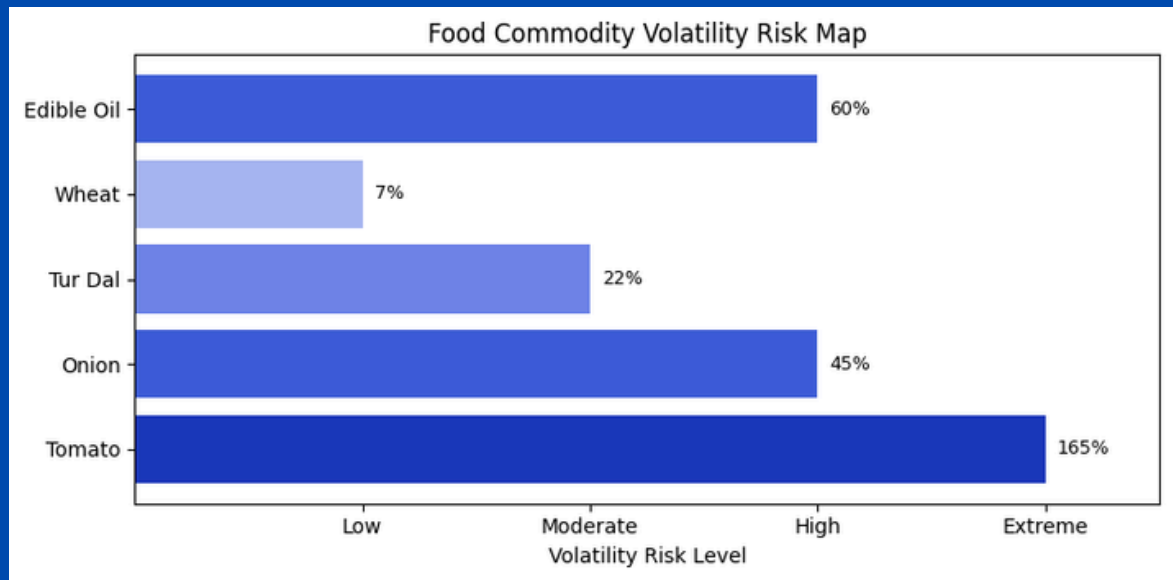
3. Emergency imports and logistics are trigger-based, preventing routine trade distortions.



APPENDIX

VOLATILITY CONCENTRATION ACROSS COMMODITIES

Commodity	Peak Price (₹/kg)	YoY Change
Tomato	200	165%
Onion	90	45%
Tur Dal	160	22%
Wheat	MSP-linked	7%
Edible Oil	Import-driven	High volatility



Food inflation risk is concentrated in a few commodities—enabling targeted, low-cost volatility management.

WHY TRADITIONAL PRICE CONTROLS BACKFIRE

Policy Tool	Intended Goal	Actual Side-Effect
MSP hikes	Farmer protection	Fiscal stress, surplus
Export bans	Consumer relief	Farmer price crash
Stock limits	Anti-hoarding	Reduced storage
Buffer release	Price control	Wastage, delays

